Application research on construction safety management of building engineering based on computer vision technology

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Abstract

With the rise of artificial intelligence, the application of computer vision technology in the construction safety management of construction engineering is increasingly extensive. Exploring the application of computer vision technology in the construction safety management is not only an upgrade of the existing safety management mode, but also provides theoretical contributions and technical support for the realization of future intelligent construction. Therefore, this study discusses the application of computer vision technology in the construction safety management of construction projects by constructing a construction engineering safety control framework. This framework evaluates the construction safety situation reasonably and quantitatively through the processing process of image acquisition and preprocessing, image recognition, data analysis and control system on the construction site. Realize the visual management of construction site personnel, construction equipment, construction materials, construction standards and construction environment, and improve the safety management effect of construction site.

CCS Concepts

• **Applied computing** → Physical sciences and engineering; Engineering; Computer-aided design.

Keywords

Computer vision technology, construction safety management, control framework, construction engineering

ACM Reference Format:
Yuanyuan Han. 2024. Application research on construction safety management of building engineering based on computer vision technology. In *2024 5th International Conference on Computer Science and Management Technology (ICCSMT 2024), October 18–20, 2024, Xiamen, China.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3708036.3708052

1 Introduction

In the field of construction engineering, construction safety management has always occupied a pivotal position [1]. However, the traditional construction management system often relies too much on the personal ability and experience of managers, and lacks systematic and standardized construction processes and codes of conduct, resulting in many uncertainties in construction safety [2]. In view of this, the deep integration of artificial intelligence technology into the construction safety management system of construction engineering and the construction of an intelligent and safe construction management model has become the focus of common attention of all sectors of society [3]. Among them, as a core branch of artificial intelligence, computer vision, with its excellent image and video capture, analysis and processing capabilities, has shown great potential in the application scenarios of construction safety management of construction engineering [4]. Cheng et al. collect and process the location data of workers and construction equipment in real time through visual inspection technology, so as to achieve the safety management of construction workers [5]. Cui Bing et al proposed a set of fatigue operation detection system for construction machinery operators, which extracts facial feature information based on computer vision technology, and then evaluates the fatigue state of operators through algorithms [6]. These studies all show that computer vision technology can make up for the limitation of traditional manpower management mode and inject new technical vitality into construction engineering safety management. However, it is worth noting that although the application of computer vision technology in the safety management of construction engineering has broad prospects, the systematic review of its application research is slightly insufficient. Therefore, in-depth research and exploration of construction engineering safety management technology based on computer vision not only has important theoretical significance, but also has far-reaching practical value for improving the level of construction engineering safety management and ensuring the safety of construction personnel.

In view of this, this study aims to systematically discuss the control framework and specific application scenarios of computer vision technology in construction engineering safety management, and further enrich and expand the application research of computer vision technology in the field of construction engineering by building a comprehensive and scientific theoretical system and practical model, so as to provide strong support for improving the level of safety management in the industry.

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ICCSMT 2024, October 18–20, 2024, Xiamen, China
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ACM ISBN 979-8-4007-0999-9/24/10
https://doi.org/10.1145/3708036.3708052

2 Construction engineering safety control framework based on computer vision technology

Building engineering safety management and control framework based on computer vision technology includes image acquisition and preprocessing, image recognition, data analysis and control system modules (Figure 1). The image acquisition and preprocessing module is responsible for efficiently capturing high-definition image or video data of the construction site, and using advanced image processing technology to preprocess it. Subsequently, the image recognition module uses algorithms such as deep learning and pattern recognition to accurately identify the pre-processed image content. The data analysis module further quantifies the identification results, evaluates the safety status of construction activities, and predicts possible safety risks. Finally, the control system module, as the decision-making and execution center of the entire framework, sends an alert to relevant managers immediately when specific security problems are identified, and provides detailed problem description and solution suggestions, according to the safety information generated by data analysis results. Through a highly integrated technical path, this framework realizes the comprehensive and real-time monitoring and management of the construction site.

2.1 Image acquisition and preprocessing

In construction sites, a variety of image acquisition equipment should be deployed according to the environmental characteristics to meet the needs of image acquisition in different scenes. First of all, in the daily construction occasions with a wide view, multi-directional surveillance cameras should be installed for real-time monitoring of the construction site, recording personnel activities, material stacking, mechanical equipment operation and so on. Secondly, it should be equipped with UAV equipment with high-altitude shooting capability, which can obtain panoramic images and high-definition video of the construction site, so as to be used for safety inspection and terrain mapping of the construction site. In special environments, such as night construction monitoring, hidden area inspection, etc., it is necessary to equip infrared cameras, night vision cameras and other professional equipment to complete image acquisition. In addition, in the detection of some large construction equipment such as tower cranes, it is necessary to use high point video surveillance to avoid the existence of visual blind areas and cause security risks. Therefore, in the practice of construction site image acquisition, it is necessary to comprehensively adopt a variety of equipment and technical means to build a multi-level and multi-dimensional image acquisition system. Subsequently, the acquired images are preprocessed as a key pre-step in the image analysis process. The preprocessing aims to improve the signal-to-noise ratio and overall quality of the images by adjusting visual attributes such as image contrast, brightness and sharpening, as well as applying noise suppression technology to lay a solid foundation for subsequent higher-level image recognition and analysis tasks. This process not only improves the availability and reliability of image data, but also enhances the accuracy and effectiveness of subsequent analysis results.

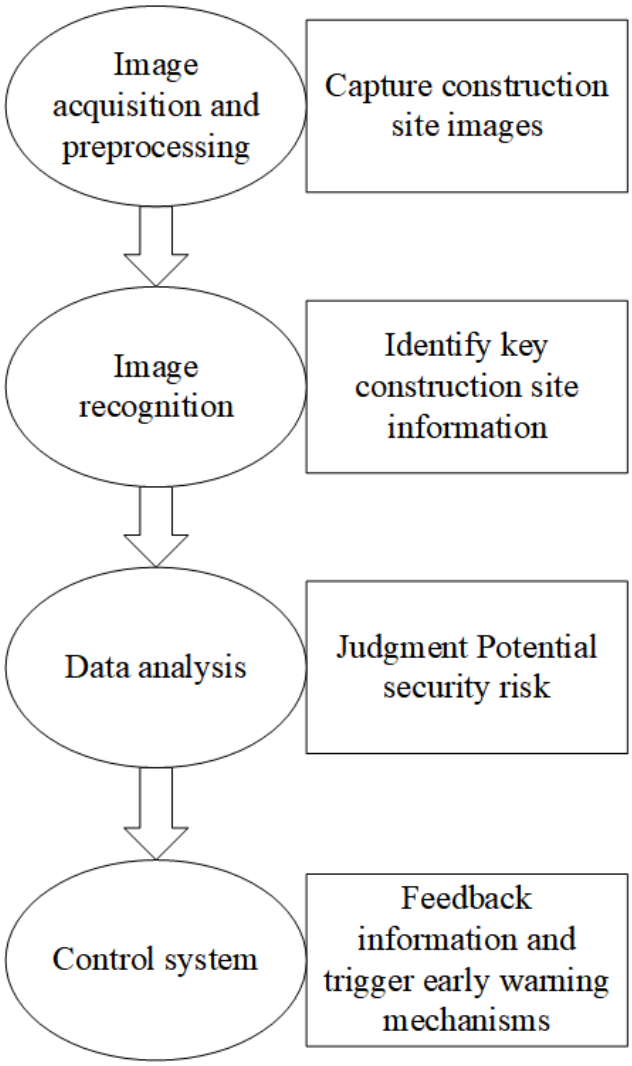

Figure 1: System management and control framework.

2.2 Image recognition

Image recognition is the core module of construction engineering safety control frame, which can identify the key information of the input image. Multiple tasks such as image classification, object detection and image semantic segmentation, which are included in computer vision technology, play an important role in the image processing of construction engineering. Among them, the object detection technology realizes the accurate location and recognition of one or more object objects through the in-depth analysis of image or video sequence. The development of object detection technology has experienced the evolution process from traditional object detection algorithm to deep learning-based object detection algorithm [7]. With the development of deep learning technology, SPP-Net, R-CNN, Fast R-CNN, Faster R-CNN, SSD and YOLO series algorithms have been introduced successively, and the performance of target detection algorithms has been significantly improved. Among them, Faster R-CNN has become a new milestone in the

field of target detection based on the optimization of R-CNN and Fast R-CNN [8].

Faster R-CNN is applied to identify unsafe factors in the construction phase of construction projects, such as the identification of hard hats worn by construction workers [9]. This technical process is shown in Figure 2, and generally follows the following process and formula [10]. The convolutional neural network will first be used to extract features from the input images, and the original images will be converted into feature maps with rich semantic information to form feature maps. It can be expressed by the formula (1):

$$F = CNN(I) \tag{1}$$

Where, I indicates Image and F indicates feature maps. These feature maps will be used as input for the subsequent processing flow. Then, Region Proposal Network (RPN), the core component of Faster R-CNN, is used to generate region proposals of objects from feature maps of input images in real time. On the basis of feature maps, RPN network uses softmax to determine whether Anchor is Positive or Negative, and also uses Bounding Box Regression to modify Anchor. Thus, more accurate region proposals can be obtained. The generation of region proposals can be expressed by the formula (2):

$$R = RPN(F) \tag{2}$$

Where R represents a series of region proposals that will be used as input to the Fast R-CNN network for further classification and regression. The main thing is to send a series of region proposals are sent to the Fast R-CNN detector, which uses Roi Pooling technology to collect the input feature maps and proposals. After synthesizing these information, the proposal feature maps are extracted. Classification network uses the proposal feature maps to calculate the category of the proposal, and then uses Bounding box regression to obtain the final accurate location of the detection box and output the location information of the target. This process is expressed in the formula (3):

$$O = FC(R, F) \tag{3}$$

Where O represents the category and location information of the output target. The object detection technology can identify and classify the images of the construction site, which provides an important basis for the processing of safety information, and helps the project managers to coordinate the safety management better.

Among the numerous target detection algorithms, the algorithms used in different scenarios are also different. Therefore, in the field of target detection, Precision, Recall, Average Precision (AP) and mean Average Precision (mAP) are commonly used to evaluate the accuracy and validity of the model. If the image of the construction site is input into the trained model for measurement, a prediction box will appear based on the position of the object, which is a positive example box. When the positive example box identifies the true sample, then the positive example box is TP; when the positive example box identifies the wrong sample, then the positive example box is FP. Conversely, the positive sample is incorrectly identified as the negative sample is FN.

Precision can measure the number of real positive samples among all identified positive samples, which represents the accuracy rate of model detection. The calculation method is shown

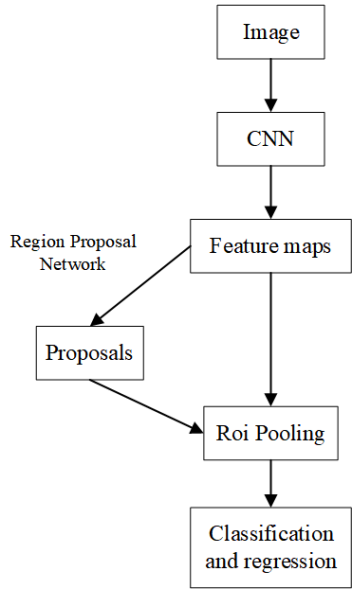

Figure 2: Faster R-CNN target detection flow chart.

in formula (4):

$$P = \frac{TP}{TP + FP} \tag{4}$$

Recall can measure the proportion of all positive samples correctly identified as positive samples, which represents the comprehensiveness of model detection. The calculation method is shown in formula (5):

$$R = \frac{TP}{TP + FN} \tag{5}$$

Average Precision (AP) is the area below precision-recall curve, which represents the comprehensive performance of target detection algorithms under different recall rates. The calculation method is shown in formula (6):

$$AP = \frac{\sum_{k=1}^{m} (P(k) * r(k))}{|R(n)|} \tag{6}$$

mean Average Precision (mAP) is the average of AP values of all categories, which can be used to evaluate the overall performance of multi-class target detection algorithms. The calculation method is as follows (7):

$$mAP = \frac{\sum_{n=1}^{N} AP(n)}{N} \tag{7}$$

By measuring the detection effect of the model through evaluation indicators, the suitable algorithm model can be applied according to the characteristics of the scene, so as to improve the feasibility and accuracy of the construction safety management of construction engineering based on computer vision technology.

2.3 Data analysis

Based on the image recognition module, the data analysis module can integrate, summarize, deeply analyze and analyze the image information captured and recognized by computer vision technology

in the statistics department. Through the construction of image database and hazard database, the comprehensive and real-time tracking and evaluation of construction personnel behavior, material status, mechanical equipment operation, construction process execution and environmental conditions can be realized, the future development trend of the construction site can be predicted, and the safety risk level under the current situation can be evaluated according to it, and the management can be provided with accurate and timely decision support. This mechanism significantly enhances the early warning and prevention capabilities in the safety management of construction projects, making safety control measures pre-emptive and accurate, and effectively ensuring the safety and stability of the construction site.

2.4 Control system

The core function of the control system, as the terminal executive module of the construction engineering safety management and control framework, is to integrate and respond to the project safety state information transmitted by the image recognition and data analysis module. With a high degree of intelligence and automation, the system can monitor and analyze feedback data in real time to accurately identify emergency safety incidents on the construction site. Once a potential or existing safety risk is detected, the control system will quickly start the alarm mechanism to send a clear and timely warning to all personnel on the construction site to ensure that relevant personnel can quickly notice the safety hazards and take effective measures to intervene, so as to effectively curb the deterioration of the safety situation and avoid or reduce potential economic losses and casualties. In addition, the control system provides a powerful monitoring and decision support platform for project managers. Through the system integrated monitoring equipment, managers can intuitively and comprehensively grasp the real-time dynamics of the construction site, including personnel activities, equipment status, environmental changes and other key information. On this basis, once any safety problems or potential defects are found, managers can quickly formulate and adjust safety rules and countermeasures according to the data and analysis results provided by the system, so as to achieve accurate management and effective prevention and control of safety risks, and ensure the smooth progress of construction projects in a safe and efficient environment.

3 Application scenario of computer vision technology in construction engineering safety management

3.1 Construction personnel management

Computer vision technology strengthens the safety management of construction site personnel. On the one hand, face recognition technology can verify the identity of personnel entering the construction site [11], avoiding the intrusion of unrelated personnel and ensuring that there will be no accidental injury accidents on the construction site. On the other hand, computer vision technology can also identify and analyze unsafe behaviors of construction workers [12], such as wearing safety equipment, operating posture, working hours, personnel gathering, etc. If unsafe behavior of construction personnel is found, computer vision technology can capture the corresponding image, and send an early warning signal to relevant personnel in time through accurately processed image data and algorithm analysis results, so as to promote the rapid implementation of safety management measures and improve the timeliness and reliability of construction project safety management.

3.2 Construction equipment management

Computer vision technology can monitor the operating status of equipment and display abnormal conditions in mechanical operation, such as excessively high equipment temperature, abnormal vibration or improper operation. Other large construction machinery, such as tower cranes, cranes and other high-risk operation equipment, their operation process is often accompanied by a complex visual environment, including the problem of visual blind areas, which become the high incidence of construction accidents, easily lead to accidental collisions between machinery and personnel, other equipment. The introduction of computer vision technology can realize the effective monitoring and predictive analysis of these blind areas, and improve the level of safety management and work efficiency of construction sites.

3.3 Construction materials management

Construction site construction materials management is also a key link. In view of the fact that the construction site often has a broad open space, this feature has led to the increase of the arbitrariness of the construction personnel in the material layout to a certain extent, forming a potential safety hazard. The disorderly placement of some building materials, such as scaffolding and construction ladder, foreshadows the occurrence of safety accidents. The computer vision system can assist the engineering management team to start the early warning mechanism, timely identify and report potential material placement risks, so as to effectively avoid improper material layout problems caused by human negligence, and significantly improve the safety management level of the construction site.

3.4 Construction standard management

In the field of construction, a series of safety standards are usually formulated, but many safety measures have not been properly implemented due to the uneven knowledge reserve of operators and the inadequate supervision of managers. For example, the edge area of stairs needs to be protected; The construction standard of safety passageway should set up double-layer protection. Once these construction standards are ignored, the safety consequences can only be borne by the construction personnel. Computer vision technology can replace human labor to check whether these construction processes meet safety standards and improve the safety of the construction site.

3.5 Construction environment management

Computer vision technology can monitor the environmental conditions of the engineering site to achieve real-time monitoring and evaluation of dust concentration, noise level and air quality to ensure compliance and safety of the working environment. And the system can continuously monitor the weather conditions, early

warning of extreme weather events, such as storms, high temperatures, etc., so that the engineering team can take necessary preventive measures to reduce losses and accidents caused by weather. With the ability of computer vision technology, the monitoring of environmental conditions on the construction site has reached an unprecedented level of refinement, providing a strong guarantee for the compliance and safety of the working environment.

4 Conclusion

Artificial intelligence-driven computer vision technology has shown broad application prospects in the field of construction safety, and its core value is to build an integrated and intelligent security control framework. Through a series of precise steps, including image acquisition and preprocessing, image recognition, data analysis and immediate response of the control system, the framework realizes all-round and multi-dimensional quantitative evaluation of construction safety. This process not only significantly improves the accuracy and timeliness of risk assessment, but also further activates the risk early warning mechanism, enabling security management to stay ahead of potential threats and effectively curb the occurrence of security accidents. Through visual management means, computer vision technology realizes the comprehensive monitoring and dynamic adjustment of construction site personnel behavior, construction equipment status, construction material layout, construction standards implementation and construction environmental conditions, which greatly enhances the safety management level and emergency response ability of the construction site. Computer vision technology has brought construction engineering safety management into a new era of more intelligent, refined and efficient, providing a strong technical guarantee for the sustainable development of the industry and the safety and health of workers.

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